

Arya Aji

Civil Engineering | Hydraulic Engineering | BIM & Sustainable Infrastructure

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PROFESSIONAL SUMMARY

Master's student in Civil Engineering at Technical University of Munich with interdisciplinary experience in hydraulic engineering, sustainable infrastructure, BIM, and construction management. Hands-on experience in dam design (DIN 19700), urban energy systems, and IFC-based model checking, complemented by strong skills in physical modeling, engineering analysis, and digital tools for real-world infrastructure challenges. Motivated to contribute to innovative and data-driven engineering solutions in infrastructure and the built environment.

TECHNICAL PROFILE

BIM and Digital: Revit, Solibri, IFC modeling, Model checking.

Design Simulation and Planning: AutoCAD, Revit, STAAD Pro (basic), Primavera P6, LCA.

Programming Skills: MATLAB.

GIS & Data Analysis: QGIS, MS Excel, Quantity Surveying, Mapping.

Surveying And Field Work: DGPS, Total Station.

Hydraulics and Water Systems: Hydraulic analysis, Flow regimes, Sediment transport, Hydraulic modeling.

WORK EXPERIENCE

Site Engineer, *Fine Form Architects*

07/2023 – 02/2024

Pandalam, India

- Supervised on-site construction activities, ensuring compliance with engineering drawings, specifications, and safety standards.
- Prepared quantity take-offs and supported cost estimation, resource planning, and technical documentation.
- Assisted in site coordination and progress tracking to support efficient project execution.

EDUCATION

MASTERS OF SCIENCE, CIVIL ENGINEERING,

04/2024 – Present

TECHNICAL UNIVERSITY OF MUNICH (TUM), Germany

Focus Areas: Sustainable Infrastructure, Hydraulic Engineering, Real Estate, BIM, Urban Systems.

BACHELOR OF TECHNOLOGY, CIVIL ENGINEERING,

07/2019 – 06/2023

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY, Kerala, India

Final grade: 2.3

Focus Area: Structural Analysis, Fluid Mechanics, Geotechnical Engineering, Transportation Engineering.

PROJECTS

RCC Dam Design – Thuringia, Germany (TUM)

04/2025 – 07/2025

- Designed a reinforced concrete dam (DIN 19700) for water supply and flood protection, integrating the existing Vorsperre Deesbach dam into the reservoir system.
- Developed reservoir planning and zoning concepts supported by hydraulic analysis and GIS-based site assessment.

Urban Systems & Sustainability Analysis – Paul-Gerhardt-Allee, Munich (TUM)

04/2025 – 07/2025

- Analyzed the study area across mobility, green & blue infrastructure, energy, and waste systems to evaluate overall urban performance.
- Led the energy-focused analysis, assessing demand and efficiency, and contributed to integrated sustainability strategies for improved resource efficiency and quality of life.

BIM Modeling & IFC Model Checking – Office Building (TUM)

04/2025 – 07/2025

- Developed an IFC-based C-shaped office building model (40 × 16 m, height 4.5 m), including structural and architectural elements (columns, beams, slabs, windows, doors, and glazed façade), structured using IfcProject, IfcSite, IfcBuilding, Storeys, and Spaces with semantic attributes (e.g., LoadBearing, IsExternal).
- Performed rule-based model checking in Solibri for door clearance, window ratio, space accessibility, and area compliance, identifying and resolving property set inconsistencies and classification errors.

City Indicators & Urban Systems Analysis – Alexandria, Egypt (TUM)

10/2024 – 03/2025

- Analyzed urban systems including mobility, green & blue infrastructure, circularity, and energy
- Led the energy analysis by estimating demand and developing basic scenarios, contributing to sustainable planning strategies.

Sustainable Construction Materials Development (Bachelor Project)

07/2022 – 06/2023

- Developed eco-friendly clay blocks using red mud, rice husk, and water hyacinth, optimizing mix design to improve structural performance and durability.
- Conducted material testing (compressive strength, water absorption, hardness, soundness) following relevant Indian Standards (IS codes), applying structural and geotechnical evaluation methods.

PRACTICAL WORK

HYDRAULIC MODEL ENGINEERING– OBERNACH RESERACH STATION (TUM)

03/2026 – 04/2026

- Conducted physical modeling on scaled hydraulic systems (Moosburg, Landquart, Baldeysee) to analyze flow behavior, hydraulic jump formation, energy dissipation, and sediment transport processes
- Evaluated system performance under varying discharge and operating conditions through experimental observation and structured data analysis

HYDRAULICS LABORATORY– OBERNACH (TUM)

05/2025 – 05/2025

- Performed laboratory experiments on weir hydraulics, spillways, stilling basins, bottom outlets, cavitation, and surge tanks to understand hydraulic behavior and system response
- Operated measurement instruments, collected flow data, and analyzed key hydraulic parameters to assess performance and efficiency

CONTOUR CREATION & MAP CREATION (BACHELOR'S)

10/2021 – 10/2021

- Conducted contour surveying over a 5000 m² area using DGPS-enabled Total Station (Trimble M3) for accurate spatial data collection
- Processed and interpreted XYZ data to generate detailed 2D and 3D terrain models using Surfer software

LANGUAGE SKILLS

English (C1)**German (A2)****Malayalam**
Mother Tongue**Hindi****CERTIFICATIONS**

EU Taxonomy and Sustainable Finance, Munich-05/2025**Geotechnical Investigation and Building Construction, Madeckal Constructions, Muvattupuzha, Kerala- 05/2021****Auto CAD-Civil, Goldmine Global Delivery Service, Kannur, Kerala- 04/2021****PERSONAL INTEREST**

Exploring new technologies, Outdoor activities, Travelling

ACCOMPLISHMENTS

NSS Volunteer (2019-23), Served as University Union Representative (2020-21), College Chair Person (2022-23)